ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: S. Ivan Sajiv

Roll No: 26CS352

Date:

Subject Name:

DEPARTMENT OF HUMANITIES & SCIENCES • AFFILIATED TO ANNA UNIVERSITY • CHENNAI

Self Introduction: Written Script and One-Page Profile

S. Ivan Sajiv

B.E. Computer Science and Engineering | First Year

Personal Profile
Name: S. Ivan Sajiv
Education: B.E. Computer Science and Engineering
Institution: St. Joseph's College of Engineering
Career Interest: Technology, Software Development and Cloud DevOps

Professional Summary

I consider myself a curious and ambitious individual who enjoys learning new things and exploring different possibilities. I believe learning should extend beyond classrooms and textbooks through practical experiences, creativity and experimentation.

I enjoy understanding how things work, identifying problems and finding ways to solve them. Every challenge is an opportunity to improve my knowledge, strengthen my abilities and become more confident. I want to keep exploring new areas and developing skills that will be useful in the future.

Introduction

Good morning everyone,

My name is Ivan Sajiv, and I am pursuing my first year of Computer Science and Engineering at St. Joseph's College of Engineering. I chose this field because technology influences almost every part of modern life. From mobile applications to artificial intelligence, it continues to change how people communicate, work and solve problems. I am interested in understanding the ideas and processes behind these developments.

I see college as more than a place to attend classes and earn a degree. It is an opportunity to discover my interests, improve my personality, gain practical knowledge and prepare for future responsibilities. I look forward to exploring different areas of computer science and identifying the field that interests me most.

Self Introduction: Written Script and One-Page Profile (Continued)

Personal Qualities and Strengths

I aim to strengthen my adaptability, problem-solving ability, consistency and communication skills. I believe a willingness to learn is one of the most important qualities a student can develop. Teamwork also matters because working with different people helps us exchange ideas, understand other perspectives and learn beyond what we might discover individually. Accepting feedback and learning from mistakes are essential for growth.

Technical Interests

I am interested in programming, software development and emerging technologies. I want to improve my understanding of programming languages and use them to build useful applications and solve real-world problems. I am also interested in artificial intelligence, cybersecurity and other developing areas of computer science. My current priority is to build a strong foundation and gain practical experience.

Skills

- Willingness to learn new technologies
- Problem-solving and logical thinking
- Adaptability and time management
- Communication and teamwork
- Interest in programming and technical learning

Education and Experience

Current Education: Bachelor of Engineering – Computer Science and Engineering
Institution: St. Joseph's College of Engineering
Academic Level: First Year

As I am at the beginning of my engineering journey, I am focused on gaining experience through academic projects, technical activities, workshops and collaborative learning. Projects: Digitalizing old patta documents and Expense Tracker. Certification: Azure Fundamentals (AZ-900) – in progress.

Future Goals

My immediate goal is to develop a strong foundation in computer science, improve my programming abilities and gain practical knowledge beyond the classroom. In the long term, I aspire to establish myself in the technology industry as a skilled professional who contributes to meaningful projects.

Closing Statement

I believe success is built through continuous learning, patience, effort and the willingness to improve. I still have much to learn, but I am excited about the opportunities ahead and determined to make the most of my college journey.

Thank you for giving me this opportunity to introduce myself.

Impressive Personality: Biographical Note

Shubham Londhe

Shubham Londhe is a Developer Advocate, Cloud and DevOps educator, and Platform Engineering strategist based in Pune, Maharashtra. His work focuses on making cloud infrastructure easier to understand and helping aspiring engineers learn through practical projects. He believes modern cloud architecture should be accessible, automated and grounded in real implementation rather than theory alone.

With more than nine years of experience in software engineering, application development and cloud automation, Londhe has worked across areas including Linux administration, container orchestration, CI/CD and cloud services. His technical experience includes Python, AWS, Docker, Kubernetes and Terraform, which he has used to build and scale production environments.

He currently works as a Developer Advocate at Amazon Web Services (AWS) and is also the founder and educator behind TrainWithShubham. Through these roles, he supports students and early-career professionals who want to enter cloud and platform engineering. His learning initiatives have reportedly helped more than 10,000 learners move towards careers in these fields.

One of his notable contributions is the open-source #90DaysOfDevOps initiative on GitHub. It provides a structured, project-based learning path that begins with Linux shell basics and Git version control, then progresses to multi-container Kubernetes deployments and Infrastructure as Code pipelines.

Beyond technical teaching, Londhe is active in developer communities, technical forums and learning events. Through workshops, technical explanations and open-source resources, he works to reduce barriers to learning and improve industry readiness among engineering students. His journey demonstrates the value of practical learning, community contribution and sharing technical knowledge.

Summary of Lecture Notes

NPTEL: Enhancing Soft Skills and Personality – Prof. T. Ravichandran, IIT Kanpur

Week 1: Course Introduction and Growth Mindset

The opening lectures revisit soft-skill areas such as self-management, stress, habits, listening, communication, presentations and trust. The main focus is mindset: how people interpret situations and respond to difficulties. The shoe-salesperson example shows that the same situation may be seen either as a problem or as an opportunity. Drawing on Carol Dweck's ideas, a fixed mindset avoids challenges and criticism, while a growth mindset treats mistakes as feedback and learning as a lifelong process. The lectures encourage curiosity, openness, resilience against negative stereotypes and a sense of belonging. The Hawthorne effect illustrates how attention and belonging can improve motivation.

Week 2: Time Management, Procrastination and Assertiveness

Rather than controlling time itself, people need to manage their habits and priorities. The lectures distinguish monochronic behaviour, which follows fixed schedules, from polychronic behaviour, which is more flexible. Students are encouraged to record how time is spent, identify distractions and use productive periods wisely. Covey's urgent–important matrix and the Pareto principle help prioritize tasks. Procrastination may arise from fear, perfectionism, low motivation or feeling overwhelmed. Suggested methods include dividing large tasks into smaller steps, completing the hardest task first, using the one-minute Kaizen approach, keeping the workspace organized and rewarding progress. Assertiveness is introduced as the ability to say no when necessary.

Week 3: Assertiveness, Anger and Positive Thinking

The lectures compare passive, assertive and aggressive behaviour. Assertiveness means expressing one's needs confidently while respecting other people, with the aim of reaching a fair outcome. Anger can be managed by delaying a reaction, counting, breathing deeply, exercising, choosing silence or redirecting the emotion into constructive work. Positive thinking is developed through self-reflection, replacing limiting beliefs, visualization and repeated positive self-talk. The story of two hospital patients illustrates how perspective can influence a person's experience.

Summary of Lecture Notes (Continued)

NPTEL: Enhancing Soft Skills and Personality – Prof. T. Ravichandran, IIT Kanpur

Week 4: People Skills and Being Attractive

Good interpersonal behaviour is built through optimism, honesty, humility, politeness, reliability and giving credit to others. Traits such as gossip, aggression, excessive criticism and false promises damage relationships. The three-filter test associated with Socrates asks whether something is true, kind and useful before it is repeated. Attractiveness is discussed beyond physical appearance: it includes mental curiosity, emotional maturity, authenticity, mindfulness and helping others without expecting a reward.

Week 5: English Skills and Common Errors

English is presented as an important language for international communication and professional life. The lectures address common mistakes involving confusing word pairs, Indian English expressions, subject–verb agreement, prepositions, articles and collocations. Special attention is given to countable and uncountable nouns and choosing words accurately. Regular exposure, practice and correction are emphasized as better ways to improve than memorizing rules alone.

Week 6: Humour and Money

Humour can reduce stress, support health and make communication easier. A smile and genuine laughter can improve relationships, while sarcasm may hurt others and should be used carefully. In workplaces, appropriate humour can ease conflict and encourage creativity. The lectures also connect money habits with personality. Spending, borrowing, lending and keeping promises reveal a person's integrity. Responsible financial behaviour includes tracking income and expenses, distinguishing needs from wants, avoiding impulse purchases and building value through dependable work. Wealth is also measured by things that money cannot buy.

Summary of Lecture Notes (Continued)

NPTEL: Enhancing Soft Skills and Personality – Prof. T. Ravichandran, IIT Kanpur

Week 7: Health and Love

Health influences confidence, first impressions and professional reliability. The lectures recommend regular physical activity, balanced meals, adequate water and sufficient sleep, along with habits that support a healthy routine. Love is described as patient, constant, freely given and unconditional. Managing relationships requires understanding body language, respecting timing, allowing relationships to develop naturally and responding to rejection maturely. Personal relationships should also be kept separate from professional responsibilities.

Week 8: Ethics, Etiquette, Mind, Memory and Environment

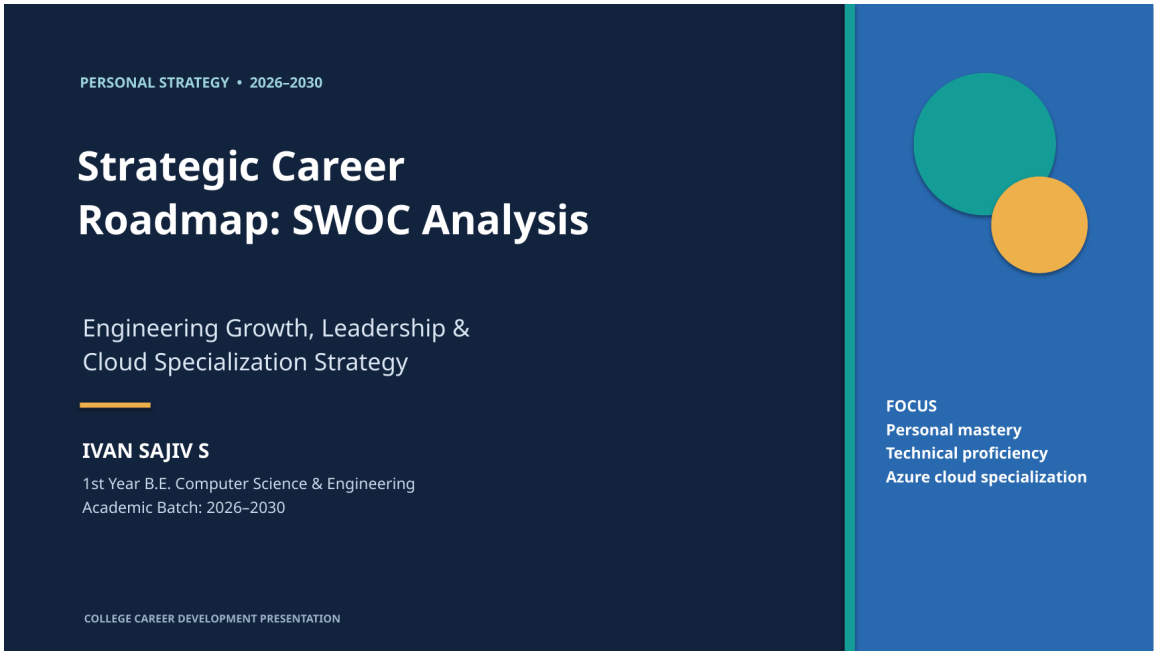
Values, ethics and etiquette shape character and guide behaviour in personal and professional settings. Professional ethics include formal rules as well as broader moral responsibilities. Good workplace etiquette involves respectful introductions, listening without interrupting, protecting confidential information, avoiding public criticism of an organization and recognizing that praise is best given publicly while correction is handled privately. The lectures also explain that the mind can be trained through positive self-talk, autosuggestion, careful thinking and fact-based decisions. Memory can be improved using association, vivid mental images, name-and-face links, number-image connections and regular mental exercise. The final topic emphasizes environmental responsibility through saving water and electricity, reducing plastic use, choosing public transport and planting trees. It encourages a shift from human-centred thinking towards care for the wider ecosystem.

Overall Reflection

The eight weeks show that professional development involves more than technical knowledge. Time management, communication, emotional control, health, ethics and environmental awareness all contribute to becoming a responsible and effective individual.

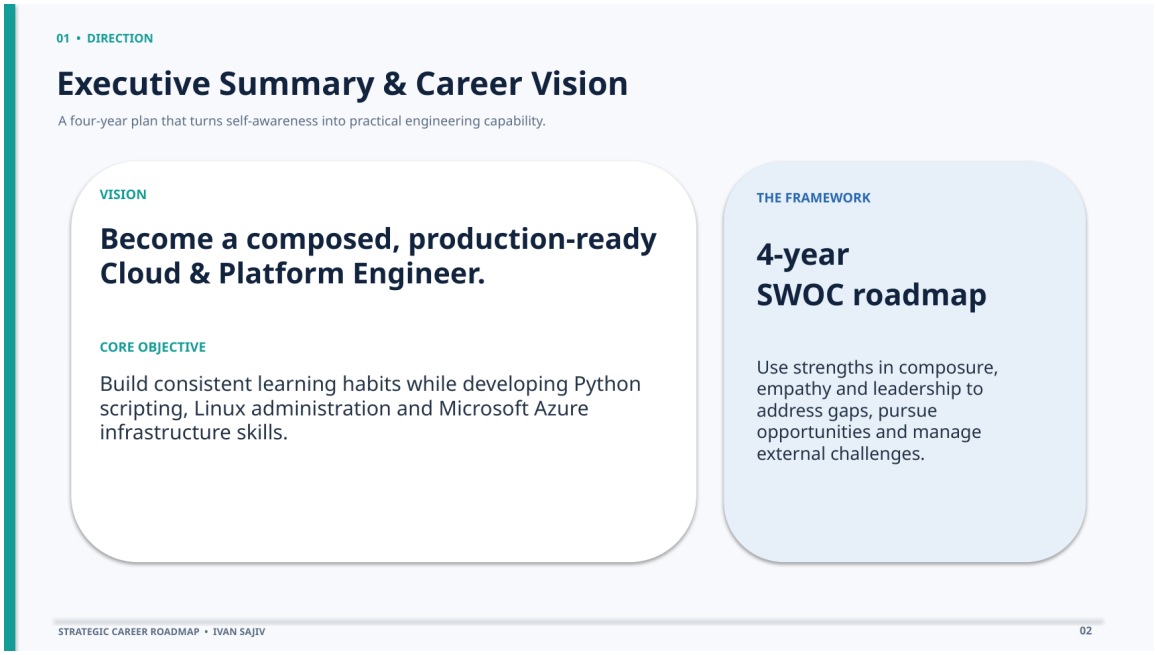
SWOC Presentation

Strategic Career Roadmap: SWOC Analysis (slides)



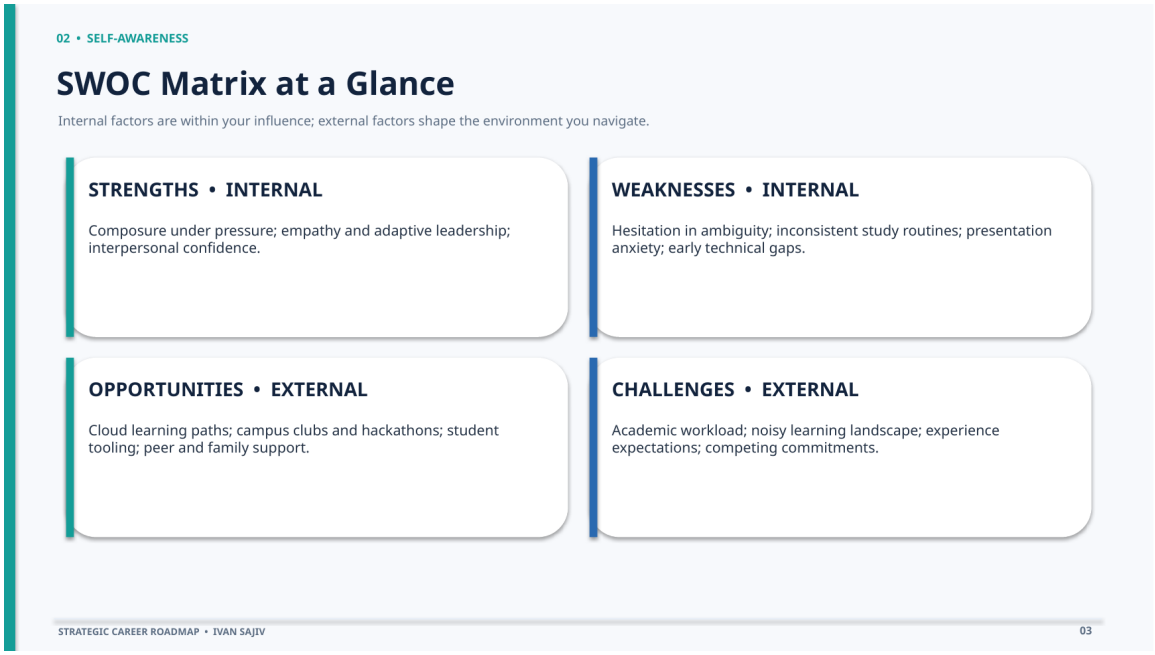
SWOC Presentation

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SWOC Presentation

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SWOC Presentation

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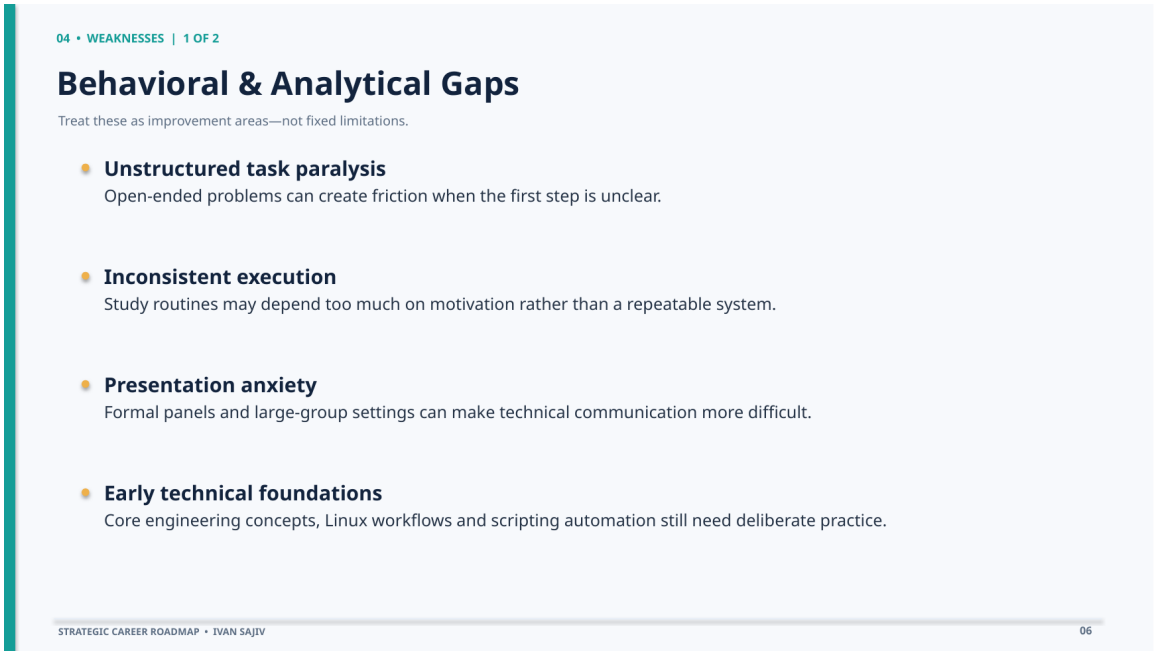
SWOC Presentation

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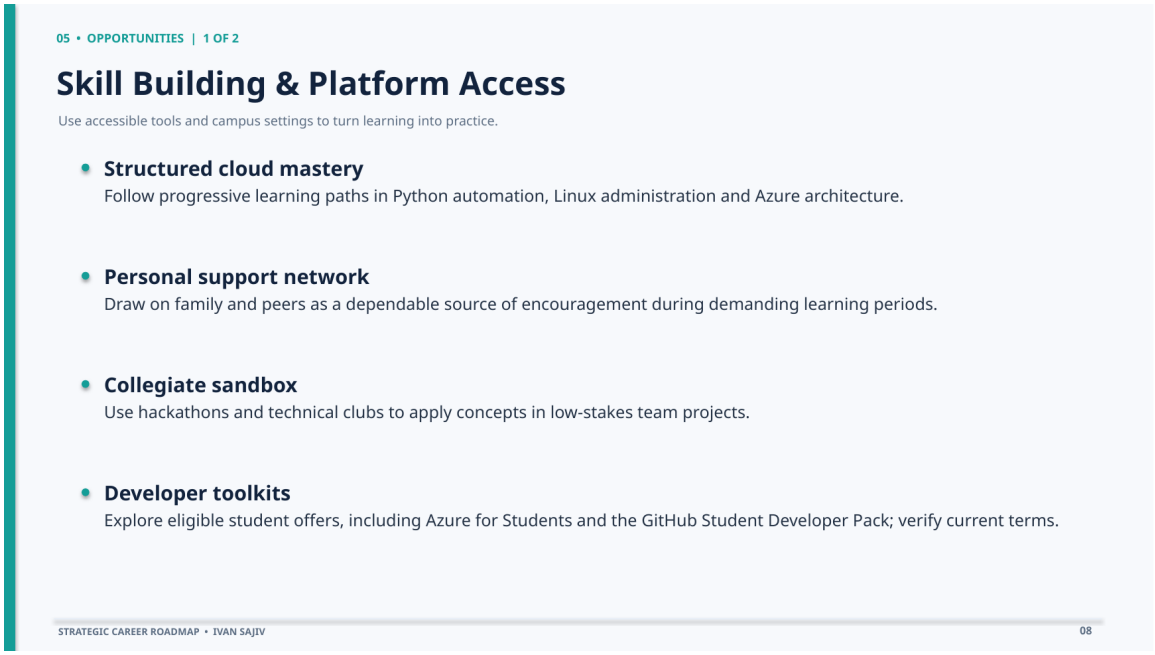
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08 • FOUR-YEAR ROADMAP

Milestones Matrix | 2026–2030

Suggested sequence: develop foundations first, then validate skills with projects and credentials.

ACADEMIC YEAR	PRIMARY FOCUS	TECHNICAL DELIVERABLE	CREDENTIAL / OUTCOME
Year 1 2026–27	Linux shell, Python basics, public speaking	Interactive portfolio hosted on Azure App Service	AZ-900; Linux+ as an optional foundation
Year 2 2027–28	Azure infrastructure, Docker, Git automation	Automated two-tier web stack using Bicep or Terraform	AZ-104 (subject to eligibility and current exam path)
Year 3 2028–29	Kubernetes, CI/CD, DevSecOps foundations	GitHub Actions pipeline for a containerized service	CKA or an Azure DevOps credential aligned to goals
Year 4 2029–30	Enterprise capstone, placement preparation	Multi-region recovery and cloud monitoring design	Placement readiness; portfolio and interview evidence

Credential names and exam requirements can change; confirm current prerequisites before scheduling.

STRATEGIC CAREER ROADMAP • IVAN SAJIV 14

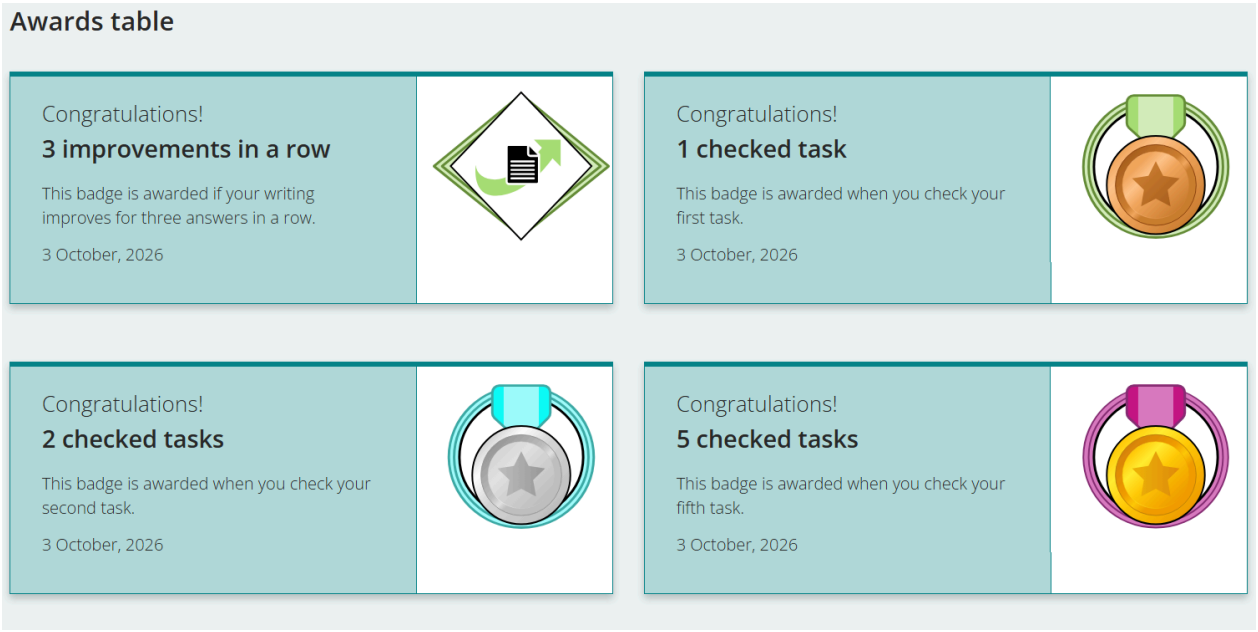
SWOC Presentation

Strategic Career Roadmap: SWOC Analysis (slides)



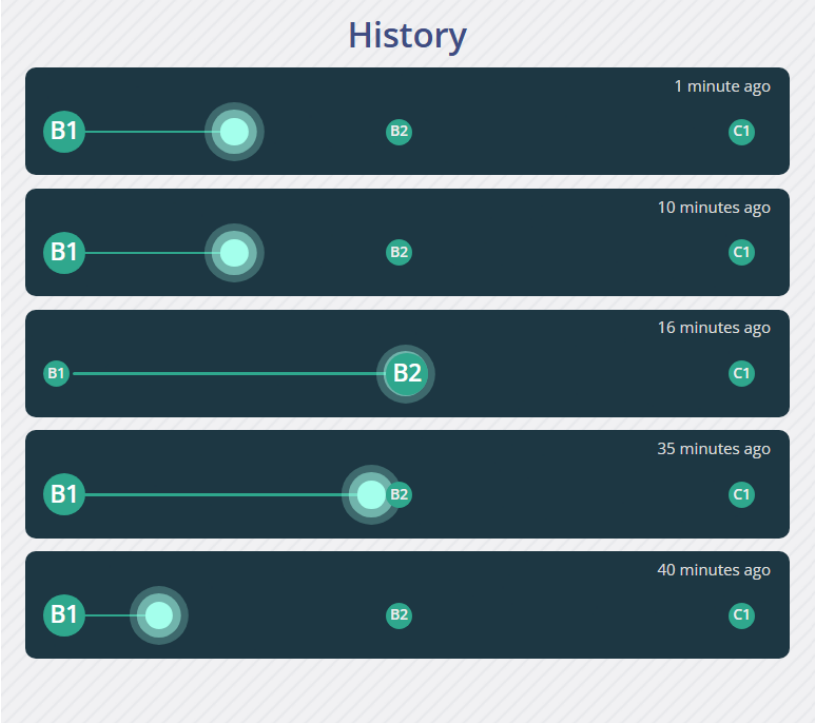
Write and Improve

Awards Table – Write & Improve (Cambridge English)

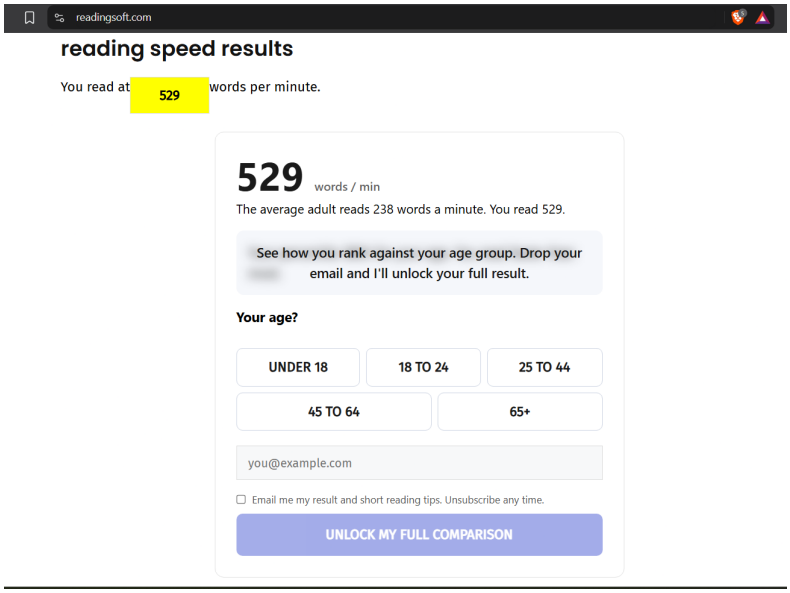


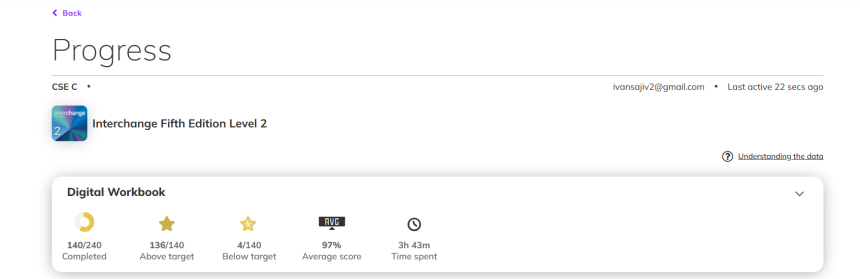
Speak and Improve

Awards Table – Speak & Improve (Cambridge English)



READING TEST





WRITE AND IMPROVE

CEFR test



PROFESSIONAL ENGLISH PRACTICE WORKSHEETS - SOLVED

COURSE CODE : EN25C01

Part A (15 × 1 = 15 Marks)

1. The automated system monitors the temperature continuously to ensure safe operation.
2. The engineers recorded the vibration readings during yesterday's trial.
3. The newly developed composite material is significantly stronger than the conventional material while maintaining a lower weight.
4. Engineers must follow safety regulations while operating heavy machinery.
5. Modern sensors can detect even minor changes in temperature.
6. The laboratory developed a new testing device that measures material strength with high precision.
7. If the temperature rises above the permissible limit, the cooling system will activate automatically.
8. If the technician had inspected the circuit earlier, the equipment would not have failed during operation.
9. The mechanics installed dual damper springs to isolate chassis resonance.
10. The research team reduced energy consumption by optimizing the motor control algorithm.
11. 50 mL of hydrochloric acid was poured into the titration flask.
12. The robotic arm moves smoothly to ensure accurate positioning of the component.
13. The project supervisor asked the engineer whether he had calibrated the pressure sensor before testing.
14. The incorrect noun is "equipments" (equipment is uncountable). Correct sentence: The laboratory purchased several new pieces of equipment for conducting material testing.
15. Because the heat sink had poor thermal paste contact, heat dissipation was worse than expected.

Part B (10 × 2 = 20 Marks)

1. The engineer calibrated the pressure sensor and verified its accuracy.
2. Portability: A laptop is more portable than a desktop computer because it is lighter, compact and runs on a built-in battery. Screen size: A desktop computer usually has a larger screen than a laptop, so it is more comfortable for long hours of work.
3. The engineer designed a new cooling system that reduces the temperature of industrial machines.
4. (a) If the battery voltage drops below the required level, the protection circuit will switch the device off automatically. (b) If the technician had checked the connections earlier, the fault would have been found and the system would not have shut down.
5. The solution was heated to 80°C before testing.
6. The project manager asked the technician whether he had inspected the pressure valves before starting the compressor.
7. (a) The engineer received a lot of feedback on the new design. ("Feedback" is uncountable, so it takes no "s" and no "many".) (b) The students received several pieces of useful advice from their teacher. ("Advice" is uncountable.)
8. (a) The technician wears safety gloves to protect his hands. (b) The cooling fan operates continuously to prevent the motor from overheating.
9. (a) The machine operates efficiently to reduce energy consumption. (b) The technician checked the electrical connections carefully before starting the machine.
10. (a) The robotic arm can lift heavy components with precision. (b) Visitors must not enter the testing area without authorization.

Part C (5 × 13 = 65 Marks)

Question 26 (a): Safe and responsible use of the internet

Safe Internet Use: Do's and Don'ts

1. Create strong, unique passwords for every account, and never share them with anyone.
2. Turn on two-factor authentication wherever it is offered.
3. Think twice before clicking links or opening attachments from unknown senders.
4. Keep your device, browser and antivirus software updated at all times.
5. Avoid sharing personal details such as your address, phone number or bank information online.
6. Use only secure websites that begin with "https" when you shop or make payments.
7. Avoid public Wi-Fi for banking or any other sensitive work; use a trusted network or a VPN instead.
8. Report cyberbullying, scams and suspicious activity to a trusted adult or the cyber crime portal right away.

Question 27 (a): A traditional art form of my region

Outline

- Selected question: 27.a
- Introduction: Bharatanatyam as the classical dance of Tamil Nadu and its ancient roots
- Body paragraph 1: History, temple tradition and revival; structure of a performance
- Body paragraph 2: Modern ways to preserve it and attract young learners
- Conclusion: Pride and participation will keep the art alive

Essay: Bharatanatyam, the Living Dance of Tamil Nadu

Tamil Nadu is home to many art forms, but few express its spirit as beautifully as Bharatanatyam. Its roots go back more than two thousand years, to the ancient Tamil epic Silappathikaram, which describes the dancer Madhavi performing on stage. For centuries the dance flourished in temples such as Thanjavur's Brihadeeswarar, where devadasis offered it as a form of worship. In the nineteenth century, the Thanjavur Quartet shaped it into the structured repertoire we see today, and in the twentieth century, Rukmini Devi Arundale revived it and gave it a respected place on the public stage.

Traditionally, a performance begins with an invocation and moves through Alarippu, Jathiswaram, Varnam and Padam, ending with a Thillana. The dancer tells stories from mythology using precise footwork, graceful body movements, mudras (hand gestures) and expressive eyes. Each movement carries meaning, so the dance is both a visual delight and a storytelling tradition. Live music, with the mridangam, violin and nattuvangam, keeps the rhythm alive.

Today, this heritage needs care. Modern initiatives can help: schools and colleges can include dance in their activity programmes, and institutions such as Kalakshetra in Chennai can offer scholarships. Online classes and recorded performances can reach children in villages, while festivals like the Chennai Margazhi season and social media can draw young audiences. Collaborations that blend classical dance with contemporary themes, such as environmental awareness, can make it relevant.

When young people learn Bharatanatyam with pride, it will continue to speak to every generation. Preserving it means protecting a living piece of our cultural identity.

Question 28 (a): Letter of request to the Librarian

Format outline

- Selected question: 28.a
- Sender's details / date: student's name, department, year, college address; 22 October 2026
- Recipient's details: The Librarian, St. Joseph's College of Engineering, OMR, Chennai - 600 119
- Subject: Request to extend library working hours during semester examinations
- Salutation: Respected Sir/Madam,
- Content: purpose, academic benefits, expected participation, arrangements

- Sign-off: Yours sincerely, name and register number

Letter

IVAN SAJIV S I Year B.E., CSE Roll No.: 26CS352 St. Joseph's College of Engineering OMR, Chennai - 600 119

22 October 2026

The Librarian St. Joseph's College of Engineering OMR, Chennai - 600 119

Subject: Request to extend library working hours during the semester examination period

Respected Sir/Madam,

I am writing on behalf of the first-year engineering students to request your kind permission to extend the library working hours during the upcoming semester examinations.

As the examinations draw near, many of us find it hard to study in hostels or at home because of noise, limited space and a lack of reference books. The library offers a quiet atmosphere, textbooks, previous question papers and journals that are very useful at this time. If the library stays open for two extra hours in the evening, from 5:00 p.m. to 7:00 p.m., students can revise properly, clear their doubts through reference material and prepare with more confidence.

We expect at least 150 to 200 students to make use of the extended hours every day. To ensure that the facilities are used properly, we request that one library assistant and a security staff member be kept on duty during this time. We also assure you that students will carry their identity cards, maintain silence, handle books with care and leave the premises on time. The Student Council will be happy to help maintain discipline.

I would be grateful if you could kindly consider our request and grant the permission. Thank you for your time and support.

Yours sincerely, [Your Name] Register No.: [Your Register Number]

Question 29 (b): Urban areas versus rural areas

Comparative matrix (selected question: 29.b)

Criteria	Urban areas	Rural areas
Employment	Many industries, IT parks, higher pay	Mostly farming and small trades, seasonal income
Cost of living	High rent, travel and food costs	Low rent, cheap food, lower wages
Education	Reputed colleges, coaching, libraries	Fewer options, long travel for higher studies
Healthcare	Multi-speciality hospitals nearby	Limited facilities, telemedicine helps
Quality of life	Convenience but pollution and stress	Fresh air and calm but fewer amenities

Drafting note: Open with a short introduction, take one paragraph per pair of criteria, and close with the idea of balanced development.

Essay: Urban Life versus Rural Life

Where we live shapes how we work, spend and relax, and cities and villages offer very different experiences.

Employment is the biggest attraction of urban life. Cities host industries, IT parks and start-ups that offer varied, better-paid jobs, while villages depend mostly on agriculture and small trades, which are seasonal and uncertain. Many young people therefore migrate to cities, hoping to build a stable career and support their families.

Cost of living is where rural areas win. Rent, food, transport and daily needs are far cheaper in a village, and families can often grow their own vegetables. However, wages in villages are also lower, so cheaper living does not always mean more savings. In cities, high rent and travel expenses can swallow much of a salary.

Education and healthcare are stronger in cities. Urban students can choose from reputed colleges, coaching centres and libraries, and patients can reach multi-speciality hospitals within minutes. In contrast, rural families may travel long distances for a college or a serious treatment, though government schemes and telemedicine are slowly narrowing the gap.

Quality of life is mixed. Villages offer fresh air, open spaces, a calm routine and close community bonds, and life moves at a slower, healthier pace. Cities offer convenience, entertainment and better infrastructure, but they also bring pollution, traffic

jams, noise and loneliness.

In conclusion, neither is perfect. Urban life offers opportunity and facilities, while rural life offers peace and affordability. The best path is balanced development, where better jobs, schools and hospitals reach villages so people can choose where to live without compromising their well-being.

Question 30 (a): Email to a school friend about the first semester

Email structure (selected question: 30.a)

- To: friend's email address
- Subject line: My first semester at engineering college
- Salutation: Hi Arun,
- Opening: ask how he is, mention long gap since last chat
- Core details: new friends, classroom activities, adjusting to college life, exam expectations
- Closing: invite him to reply, warm sign-off

Email

To: arun.kumar@example.com

Subject: My first semester at engineering college

Hi Arun,

How have you been? It feels like ages since we last talked, so I thought I would write and tell you everything about my first semester.

I was nervous on the first day, but things got better quickly. I have made some wonderful friends in my class and hostel. We study together, share notes and even have our small tea breaks outside the canteen. In class, we have a mix of theory subjects and lab sessions, and I really enjoy the Professional English lab and the programming practicals. Group projects and presentations have helped me become more confident about speaking in front of others.

Adjusting to college life was not easy at first. The timetable is tighter, the syllabus is bigger and nobody reminds you to finish assignments. But I am slowly learning to plan my time, and I have started revising a little every day instead of waiting until the last minute.

The semester examinations are coming up soon. I am a bit anxious about Mathematics and Engineering Physics, but I am preparing well and hope to score good marks. I will do my best.

Write back and tell me about your college too. Let's meet during the holidays.

Take care and keep in touch!

Your friend, IVAN SAJIV

Model I Practice Worksheet — Answer Key

Correct answers are highlighted in yellow. For open-ended writing questions, sample answers are provided; you can personalize names, places, and details.

Part A — Grammar

Exercise 1: Simple Present Tense

1. recognize
2. requires
3. transmit
4. agrees
5. form

Exercise 2: Simple Past Tense

1. conducted
2. deployed
3. opened
4. published
5. detected

Exercise 3: Comparative Adjectives

1. higher
2. safer
3. faster
4. more durable
5. lower

Exercise 4: Modals of Obligation & Necessity

1. must
2. must
3. have to
4. should
5. has to

Exercise 5: Modals of Capability & Possibility

1. can
2. could
3. can
4. may
5. can

Exercise 6: Relative Clauses

1. A civil engineer who designed the suspension cable bridge received the national infrastructure award.
2. The spectrometer, which analyzes infrared emission spectra, is mounted inside the cleanroom vacuum chamber.
3. The technical specialist whose patent on pneumatic grippers was approved was reassigned to the robotics lab.
4. This microcontroller, which features ultra-low-power sleep states, is ideal for battery-operated remote sensors.
5. The field researchers who surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

Exercise 7: If-Clauses (Type 1)

1. exceeds
2. fails
3. engages
4. rise

5. registers

Exercise 8: If-Clauses (Type 3)

1. had started
2. had lubricated
3. would have been prevented
4. would have tripped
5. had remained

Exercise 9: Infinitives of Purpose

1. The mechanics installed dual damper springs to isolate chassis resonance.
2. Technicians purge the fuel line with dry nitrogen gas to remove residual flammable vapors.
3. We introduced an optical encoder to determine spindle velocity accurately.
4. The developers wrote an automated load testing script to simulate ten thousand concurrent connections.
5. Gold plating is applied to terminal connectors to prevent atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

1. redesigning
2. replacing
3. reducing
4. cleaning
5. filtering

Exercise 11: Impersonal Passive Voice

1. It was reported that 50 mL of hydrochloric acid had been poured into the titration flask.
2. It was reported that the robotic arm had been tested under a 50 kg dynamic payload.
3. It was reported that the optical pyrometer had been calibrated prior to ignition.
4. It was observed that severe cavitation damage had occurred along the impeller blades.
5. It is stated that volatile solvents are stored in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

1. precisely
2. periodically
3. rapidly
4. firmly
5. smoothly

Exercise 13: Reported Speech (Interrogative Structures)

1. The quality manager asked the technician whether the insulation resistance test had been documented.
2. The lead architect enquired whether the server cluster could handle double the transaction throughput.
3. The field auditor asked the safety engineer whether they had inspected the gas leakage relief lines.
4. The client asked the team whether the firmware update would fix the intermittent battery drain.
5. The instructor asked the lab group why the multimeter had shown fluctuating voltage readings.

Exercise 14: Countable and Uncountable Nouns

1. The procurement officer purchased several new machines and pieces of equipment for the workshop.
2. The laboratory generated a huge amount of test data during the wind tunnel trials.
3. We need three more software licenses to complete the finite element analysis on the model.
4. The field report provided numerous valuable pieces of information regarding the soil composition.
5. All chemical apparatus and glassware must be washed thoroughly after the experiment.

Exercise 15: Comparative Adjectives (Irregular Forms)

1. worse
2. better

3. farther
4. less
5. further

Part B — Grammar and Technical Writing

Exercise 1: Simple Past Tense

1. The technician inspected the solar arrays and recorded the peak power conversion value.
2. The chemical analyzer detected trace mercury levels and sent an automated alert to the environmental dashboard.
3. Maintenance crews lubricated the heavy conveyor bearings and replaced the worn rubber belts before the morning shift.
4. The supervisory software polled field sensors every five milliseconds and archived temperature trends into a secure database.
5. The testing team applied a 500-volt dielectric pulse across the stator winding and monitored for insulation breakdown.

Exercise 2: Comparative Adjectives — Sample Answers

1. Cloud storage usually has lower initial deployment costs than on-premise physical servers. On the other hand, cloud storage offers more elastic scalability than on-premise servers.
2. Fiber-optic cables provide higher data-transfer bandwidth than copper twisted-pair cables. Fiber-optic cables are less susceptible to electromagnetic interference than copper cables.
3. Electric vehicles produce zero tailpipe emissions, whereas ICE vehicles emit exhaust gases. Electric vehicles often have a higher initial purchase price than comparable ICE vehicles.
4. Hydraulic actuators can produce greater force than many electromechanical actuators. Electromechanical actuators have a lower risk of fluid leakage because they do not use hydraulic fluid.
5. Relational SQL databases generally use a more rigid schema than document-based NoSQL databases. Some NoSQL databases support greater horizontal scalability and throughput than traditional relational systems.

Exercise 3: Relative Clauses

1. The algorithm, which was patented by the aeronautics team, analyzes microscopic cracks in turbine blades.
2. The lead chemical engineer, who noticed an abnormal pressure spike in chamber two, authorized the emergency boiler shutdown.
3. The laboratory where high-voltage insulator puncture tests are conducted houses a 400 kV impulse voltage generator.
4. We engaged an external cybersecurity consultant whose firmware vulnerability audit exposed three backdoor ports.
5. The supervisory system utilizes an artificial neural network that predicts transformer insulation life based on dissolved gas concentrations.

Exercise 4: If-Clauses — Sample Answers

1(a). If the operating temperature exceeds 85°C, the cooling system will activate automatically.

1(b). If the operator had pressed the emergency shutdown switch immediately, the equipment would have stopped before further damage occurred.

2(a). If the supply voltage drops below the 200 V threshold, the controller will disconnect the load.

2(b). If the network administrator had enabled multi-factor authentication, the account would have been harder to compromise.

3(a). If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, the ventilation system will increase airflow and an alarm will sound.

3(b). If the mechanical team had checked the bearing lubrication schedule, the bearing failure might have been avoided.

4(a). If the autonomous guided vehicle detects an obstacle within 1 meter, it will stop immediately.

4(b). If the thermal sensor had registered the runaway reaction in time, the control system would have initiated an emergency shutdown.

5(a). If the software receives an unauthenticated API token, it will reject the request.

5(b). If the civil inspectors had reviewed the core sample density tests, they might have identified the structural risk earlier.

Exercise 5: Impersonal Passive Voice

1. It was reported that the crude extract had been distilled at a constant temperature of 60°C.

2. It was reported that the cross-section of the weld seam had been polished with fine diamond paste.
3. It was reported that fifteen cast-iron pump casings had been rejected due to internal porosities.
4. It was reported that the resistance across the bridge circuit had been measured using a calibrated digital ohmmeter.
5. It was reported that the emergency aerodynamic braking flaps had been initiated at an altitude of 3,000 meters.

Exercise 6: Reported Speech

1. The safety supervisor asked the intern whether they had verified the grounding connections on the generator.
2. The plant director asked the site engineer whether the emergency effluent treatment plant had received clearance.
3. The IT security lead asked the developer why they had disabled the firewall rule on the database server.
4. The client asked the project manager whether the bridge structure could withstand Category 4 hurricane winds.
5. The research guide asked the postgraduate scholar how many sample trials they had conducted under cryogenic conditions.

Exercise 7: Countable vs. Uncountable Nouns

1(a). We need to install some new software and machinery in the CAD laboratory.
1(b). The team collected plenty of useful information from the field test.
2(a). The technician ordered ten new pieces of apparatus and modern electronic equipment for the antenna lab.
2(b). High-speed train operations require advanced transport infrastructure and dedicated electrical power.
3(a). The environmental sensor gathers extensive atmospheric data and pollution indices every minute.
3(b). The construction crew unloaded tons of sand, cement, and steel debris at the worksite.
4(a). All electrical hardware and wiring must comply with local safety legislation.
4(b). The data scientist gained interesting knowledge after running the machine-learning clustering.
5(a). Laboratory staff must clear away all glassware and acidic waste before leaving the premises.
5(b). The project failed due to a lack of proper funding and technical research.

Exercise 8: Infinitives of Purpose & Gerunds

1(a). A pressure relief valve is installed to vent excess boiler steam.
1(b). The automated script runs every midnight to archive legacy server logs.
2(a). Automotive engineers installed an intercooler to reduce the compressed-air intake temperature.
2(b). The technicians analyzed the structural stress points by placing piezoelectric strain gauges along the girder.
3(a). High-performance heatsinks feature copper heatpipes to conduct heat away from the silicon die.
3(b). The factory cut water consumption by 30% by implementing a closed-loop recycling filtration system.
4(a). Safety guards are mounted over rotating flywheel belts to prevent accidental worker contact.
4(b). The algorithm optimizes memory bandwidth by pre-fetching sequential array indices into L2 cache.
5(a). The electronics lab installed a UPS to protect servers against sudden voltage sags.
5(b). The metallurgy group enhanced surface hardness by diffusing nitrogen atoms into the steel casing.

Exercise 9: Adverbs of Manner

1(a). automatically
1(b). flawlessly
2(a). accurately
2(b). instantly
3(a). continuously
3(b). cautiously
4(a). consistently
4(b). catastrophically
5(a). periodically
5(b). smoothly

Exercise 10: Modals

1(a). can
1(b). must not

2(a). must
2(b). might
3(a). must not
3(b). could
4(a). should
4(b). must
5(a). can
5(b). must not

Part B — Topic-based Writing Samples

The worksheet includes several alternative prompts under each writing topic. The following are model responses for one prompt from each topic. Adapt details to your own college, hometown, and experience.

Topic 1: Imperative Safety Instructions — Chemistry & Materials Testing Laboratory

1. Wear a lab coat, chemical-resistant gloves, and safety goggles at all times.
2. Read the labels and safety data sheets before handling any chemical.
3. Use a fume hood when working with volatile or hazardous substances.
4. Never pipette chemicals by mouth.
5. Keep food and drinks outside the laboratory.
6. Label every container clearly and keep incompatible chemicals separate.
7. Dispose of chemical waste in the designated, labelled containers; never pour it into the sink unless authorized.
8. Clean spills immediately using the approved procedure and report accidents to the supervisor.

Topic 2: Harvest Festival and Eco-Friendly Practices

A Traditional Harvest Festival and a Greener Future

Harvest festivals are joyful occasions that bring families, farmers, and neighbours together to thank nature for a successful crop. In Tamil Nadu, Pongal is one of the most loved celebrations. Homes are cleaned and decorated with colourful kolams, and people wear traditional clothes. In many villages, families prepare sweet Pongal using newly harvested rice, milk, jaggery, and cashew nuts. The dish is allowed to boil over as a symbol of abundance and prosperity.

The festival also celebrates the hard work of farmers and the importance of cattle in agriculture. People visit relatives, share food, and take part in cultural activities. These customs strengthen community bonds and help younger generations understand their heritage.

Today, we can preserve the spirit of Pongal while reducing harm to the environment. Families can use banana leaves or reusable plates instead of disposable plastic. Decorations can be made from flowers, leaves, and natural materials. Organic waste from cooking can be composted, while leftover food can be shared safely with others. Communities can avoid loud firecrackers and use energy-efficient lights or solar-powered decorations.

Local authorities can provide clearly marked recycling bins and encourage people to walk, cycle, or use shared transport to festival events. By combining traditional values with practical eco-friendly habits, we can celebrate Pongal responsibly. A greener festival protects the environment without losing its warmth, cultural meaning, or sense of togetherness.

Topic 3: Technical Study Tour to a Hydroelectric Dam

Our class can begin the four-hour journey to a nearby hydroelectric dam by college bus at 6:30 a.m. Before departure, the faculty coordinator should confirm visitor permission, student attendance, first-aid supplies, and safety instructions. We should carry identification, water bottles, notebooks, and packed snacks.

After arriving, the group can attend a briefing on the dam’s history, water management, and electricity generation. With an authorized guide, students can observe the reservoir, intake gates, penstocks, turbines, and generator hall from designated visitor areas. The guide can explain how stored water turns turbines and how the generator converts mechanical energy into electrical energy. Students should stay behind barriers, wear the required protective equipment, and never touch operating machinery.

At lunch, we can try regional food such as rice, sambar, vegetables, and a traditional sweet from a nearby town. Buying food from local vendors supports the community and gives students a taste of regional culture. In the afternoon, students can record observations about water flow, environmental protection, and the landscape surrounding the plant. They should avoid littering and follow photography rules.

An estimated per-student budget is ₹500 for bus travel, ₹250 for meals and snacks, ₹100 for entry or administrative charges, and ₹150 for contingency expenses, totalling ₹1,000. The group can return to college by early evening and submit a short technical report comparing the dam's energy output, safety systems, and environmental benefits.

Topic 4: Formal Letter of Request — Industrial Visit

From: [Your Name], Class Representative, First-Year Engineering, [College Name], [City]

Date: [Date]

To: The Plant Manager, [Industrial Robotics and Automation Factory], [City]

Subject: Request for Permission to Visit the Industrial Robotics and Automation Plant

Respected Sir/Madam,

On behalf of our department, I kindly request permission to arrange an industrial visit to your robotics and automation facility for 50 first-year engineering students, accompanied by faculty members. The visit would help students connect classroom concepts with real industrial applications.

Our objectives are to understand industrial robot components, automated assembly lines, sensor-based control, quality inspection, and workplace safety procedures. We would be grateful for a guided tour and a short technical briefing by your engineers. We will follow all visitor rules, wear the required personal protective equipment, and remain with our faculty coordinators throughout the visit.

We are flexible regarding the date and will provide the student list and any required documents in advance. We assure you that the group will respect your staff, equipment, and operational requirements.

Thank you for considering our request. We look forward to your positive response.

Yours faithfully,

[Your Name]

Class Representative

[Department and College Name]

Topic 5: Compare and Contrast — Electric Vehicles vs. ICE Vehicles

Electric Vehicles and Conventional Vehicles: Benefits and Limitations

Electric vehicles (EVs) and internal combustion engine (ICE) vehicles both provide personal transportation, but they differ in cost, emissions, maintenance, and energy supply. EVs use batteries and electric motors, whereas ICE vehicles burn petrol or diesel to produce motion.

One major advantage of EVs is that they produce no tailpipe emissions. This can improve local air quality, especially in crowded cities. In contrast, ICE vehicles release exhaust gases while operating. EVs also generally have fewer moving parts and can cost less to maintain. Electricity may be cheaper than fuel per kilometre, although savings depend on local prices and driving habits.

However, EV batteries are expensive to manufacture and require minerals such as lithium and nickel. Battery production has an environmental footprint, and a battery may lose capacity over time. Conversely, ICE vehicles can usually refuel quickly, and petrol stations are widely available. EV charging infrastructure is growing, but charging can still be inconvenient in areas with few public chargers or for people without home charging.

EVs often have a higher initial purchase price, although incentives and lower running costs may reduce the total cost of ownership. ICE vehicles may be less expensive to buy, but fuel and maintenance costs can add up over time. In conclusion, EVs are a promising choice where charging is practical and electricity is relatively clean. ICE vehicles remain convenient in some regions. The best option depends on budget, travel distance, charging access, and environmental priorities.

Topic 6: Personal Email to a High School Teacher

Subject: An Update on My First Semester in Engineering

Dear Teacher,

I hope you are doing well. I wanted to share how my first semester at engineering college has been going. At first, adjusting to campus life and managing my own schedule was challenging, but I am gradually becoming more independent.

The laboratory sessions have been especially interesting. I have started learning how to use equipment safely, record observations, and work with classmates on small technical tasks. Being part of a team has taught me to communicate clearly and manage time better.

My end-semester examinations are approaching, so I have made a study timetable. I revise class notes daily, practise important questions, and set aside extra time for subjects I find difficult. I am also trying to balance theory preparation with

completing lab records on time.

Thank you for the guidance and encouragement you gave me in school. The study habits you taught me are still helping me now. I hope to keep improving and make you proud. Please share any advice you have for preparing effectively for my first university examinations.

Warm regards,
[Your Name]

Topic 7: Four-Day Technical Tour — Aerospace and Satellite Launch Station

Day 1: Travel to the launch-station city by train or college bus. Check into pre-booked accommodation, attend a safety briefing, and review the visit schedule. Estimated cost: transport ₹1,500, stay ₹700, food ₹400.

Day 2: Visit the authorized visitor centre and attend a presentation on satellite systems, launch vehicles, mission planning, and ground support equipment. Students should carry identification, follow photography restrictions, and stay with the group. Estimated cost: local transport ₹250, food ₹400, stay ₹700.

Day 3: Join the permitted technical tour of selected facilities, subject to official approval. Observe demonstrations or exhibits about telemetry, tracking, propulsion, and launch safety. Do not enter restricted zones or touch equipment. Record observations for the final report. Estimated cost: local transport and entry arrangements ₹300, food ₹400, stay ₹700.

Day 4: Attend a student discussion, visit the public science exhibition if time permits, and return home. Estimated cost: food ₹400 and return transport ₹1,500.

The estimated total per student is ₹7,250, including ₹3,000 for travel, ₹2,100 for three nights of accommodation, ₹1,600 for meals, ₹300 for local transport, and ₹250 for contingency. The faculty coordinator should confirm all bookings, permissions, emergency contacts, and insurance arrangements before departure. Students must wear suitable footwear, obey security instructions, keep identification visible, and follow every site safety rule.

IDEATION PROJECT

SMARTBIN 360°

An AI-Powered Smart Waste Management System for a Cleaner Campus

1. Introduction – The Idea

Waste management is one of the biggest problems in colleges, schools, public places, and cities. People often throw plastic, paper, food waste, and other materials into the same dustbin.

My idea is to develop SMARTBIN 360°, an intelligent waste management system that identifies, separates, and monitors waste using smart technology.

The main goal is to make waste disposal easier, reduce pollution, and encourage people to recycle.

2. Problem Statement

The major problems in existing waste management systems are:

- People mix different types of waste.
- Recyclable materials are wasted.
- Dustbins overflow without warning.
- Waste collection is not properly monitored.
- Colleges struggle to maintain clean surroundings.

Our challenge: How can we use technology to make waste management smarter and more efficient?

3. Proposed Solution

SMARTBIN 360° uses sensors, AI-based waste recognition, and a monitoring system to improve waste disposal.

The system includes:

1. AI Waste Detection: A camera-based system identifies the category of waste.
2. Automatic Segregation: Waste is directed into the appropriate compartment.
3. Fill-Level Sensors: Detect when a dustbin is nearly full.
4. Mobile Dashboard: Displays bin status and waste collection information.
5. Reward System: Students earn points for responsible waste disposal.

4. How Does It Work?

1. Step 1 – Waste Input

A student places waste into the smartbin.

2. Step 2 – Identification

A camera and AI system identify the waste category.

3. Step 3 – Segregation

The system directs the waste into the correct compartment.

4. Step 4 – Monitoring

Sensors measure the bin's filling level.

5. Step 5 – Notification

Cleaning staff receive an alert when collection is required.

5. Technologies Used

Technology	Purpose
Artificial Intelligence	Waste identification
IoT Sensors	Monitor bin levels
Arduino / ESP32	Control system
Camera Module	Capture waste images
Mobile Application	Display information
Cloud Database	Store waste data

6. Uniqueness of the Idea

What makes SMARTBIN 360° different?

- Combines AI and IoT in one system.
- Encourages students to recycle through rewards.
- Reduces unnecessary waste collection trips.
- Provides real-time waste monitoring.
- Can be implemented in colleges, shopping malls, hospitals, and public spaces.

7. Benefits and Social Impact

Environmental

Improves recycling and reduces pollution.

Educational

Creates awareness among students.

Economic

Reduces avoidable collection costs.

Social

Creates cleaner and healthier communities.

8. Future Scope

In the future, SMARTBIN 360° can be developed further by:

- Connecting multiple smartbins across an entire city.
- Introducing a recycling reward application.
- Using data analytics to study waste generation.
- Partnering with recycling companies.
- Developing solar-powered smartbins.

9. Conclusion

SMARTBIN 360° is not just an ordinary dustbin; it is a step towards a smarter and cleaner future.

By combining artificial intelligence, IoT, and environmental awareness, this idea can transform the way waste is managed in educational institutions and public places.

Our vision is simple: Don't just throw waste. Make every waste count.

Part A: LinkedIn Profile

S. Ivan Sajiv

Aspiring Cloud & Platform Engineer | CSE Student (Batch 2026–2030) | Python, Linux & Azure

St. Joseph's College of Engineering, Chennai

About

I'm a curious and ambitious Computer Science and Engineering student who enjoys understanding how things work, finding the problem and then solving it. I believe learning should go beyond textbooks, so I focus on practical projects, hands-on practice and steady daily habits. I'm building my foundation in Python and Linux, and my main career direction is cloud computing and DevOps, with a focus on Microsoft Azure. I'm also interested in artificial intelligence and cybersecurity. Outside academics, I have performing-arts experience in dance, which helps me stay calm and confident in front of people and work well in teams. My goal is to become a composed, production-ready Cloud and Platform Engineer.

Fun fact: I read at 529 words per minute, more than double the average adult's 238.

Focus areas: Python • Linux • Cloud fundamentals (Azure) • DevOps • Problem solving • Teamwork

Featured Projects

Digitalizing Old Patta Documents: A project to convert old land (patta) records into digital form. *Tech:* c,python

Expense Tracker: A tool to track income and spending. *Tech:* python

Education

B.E. Computer Science and Engineering, 1st Year, Batch 2026–2030

St. Joseph's College of Engineering (Autonomous), OMR, Chennai

Honors & Certifications

- CEFR B2 (Upper-Intermediate) English proficiency, Cambridge Write & Improve, October 2026
- Microsoft Azure Fundamentals (AZ-900), in progress

Skills

Languages	Operating Systems	Tools & Platforms
Python, C	Linux, Windows	Git, GitHub, Microsoft Azure (learning)

Strengths: Composure under pressure, empathy, adaptive leadership, taking feedback and improving, teamwork

Licenses & Certifications (Planned)

- AZ-900 Azure Fundamentals: Year 1
- AZ-104 Azure Administrator: Year 2 (subject to eligibility and the current exam path)
- CKA (Certified Kubernetes Administrator) or an Azure DevOps credential: Year 3

Career Roadmap

Year	Focus	Deliverable
Year 1 (2026–27)	Linux shell, Python basics, public speaking	Interactive portfolio hosted on Azure App Service
Year 2 (2027–28)	Azure infrastructure, Docker, Git automation	Automated two-tier web stack using Bicep or Terraform
Year 3 (2028–29)	Kubernetes, CI/CD, DevSecOps foundations	GitHub Actions pipeline for a containerized service
Year 4 (2029–30)	Enterprise capstone, placement preparation	Multi-region recovery and cloud monitoring design

Connect

GitHub: github.com/ivansajiv | LinkedIn: linkedin.com/in/ivansajiv

Hi there, I'm Ivan Sajiv S.

First-Year CSE Student | Aspiring Cloud & Platform Engineer | Software Development

I'm a first-year Computer Science & Engineering student at **St. Joseph's College of Engineering, Chennai**, interested in understanding how technology works and building things along the way.

About Me

- **Education:** B.E. Computer Science & Engineering - St. Joseph's College of Engineering
- **Batch:** 2026-2030
- **Career interests:** Cloud, Platform Engineering & Software Development
- **Programming:** C, Python
- **Tools:** Git & GitHub
- **Currently focused on:** Strengthening programming fundamentals, building projects and exploring cloud & software technologies
- **Interests:** Figuring out how technology works and experimenting with new things
- **Outside tech:** Dancing

Tech Stack

Languages: C, Python
Tools & Technologies: Git, GitHub, VS Code
Currently Exploring: Cloud Computing, Platform Engineering, Software Development

Projects

Intelligent Land Record Digitization & Validation System
College hackathon project focused on digitizing and validating historical land records using OCR, AI and GIS.

Simple Calculator
A simple calculator built to practice C programming fundamentals and user input handling.

Fake News Detection System
Academic assignment exploring basic fake-news detection and data processing.

Expense Tracker
A small personal project built to practice application development and expense tracking.

My Learning Journey

I'm currently focused on building strong fundamentals rather than trying to learn everything at once.

C -> Python -> Git/GitHub -> Cloud -> Platform Engineering -> Software Development
This is a work in progress.

Let's Connect

- LinkedIn: linkedin.com/in/ivan-sajiv-1b679b42a
- Email: ivansajiv2@gmail.com

Thanks for visiting my profile!